Tiger Schueler

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Education

University of Washington Tacoma — Tacoma, WA
Bachelor of Science in Computer Science & Systems, Minor in Mathematics, GPA: 3.6 — *Aug. 2025*

- CeDiploma: 25C9-NTAN-TNRJ

Technical Skills

Languages: C#, T-SQL, Oracle SQL, JavaScript, TypeScript, Python, Java
Frameworks & Platforms: .NET 10, ASP.NET Core MVC, ADO.NET, Azure SQL, React, Tailwind CSS
Developer Tools: Visual Studio 2022, Git, Claude Code, SSMS, Oracle SQL Developer
Databases: Microsoft SQL Server (Azure SQL), Oracle Database (CMiC, Primavera P6)

Experience

Data Engineer — Feb. 2026 – Present
BNBuilders — *Seattle, WA*

- Designed and built a production ETL pipeline platform from scratch in C# and .NET 10, replacing a $20,000/year third-party SaaS data sync tool (CData Sync) and fully decommissioning the legacy system within 3 months.
- Engineered high-throughput parallel chunked data pipelines using `SqlBulkCopy` to sync Oracle ERP (CMiC) and Primavera P6 into Azure SQL elastic pool destinations, processing 263M+ rows per day across 394 active jobs at 96%+ daily success rate.
- Built an atomic table swap finalization strategy using `sp_rename` that eliminated multi-hour MERGE operations on tables exceeding 500K rows, reducing finalization from 3+ hours to under 5 minutes for the largest datasets.
- Architected a split deployment with a .NET Worker Service running as a Windows Service on a VM and an ASP.NET Core MVC web UI hosted on Azure, sharing a metadata database with AES-256 encrypted connection strings, role-based access control, real-time job monitoring, request/approval workflow, and SMTP email notifications.
- Led frontend modernization of a second internal ASP.NET Core MVC application, migrating from Telerik Kendo, jQuery, and Bootstrap to a Vite + React + TypeScript stack with Tailwind CSS v4, shadcn/ui, TanStack Table v8, TanStack Query v5, and React Router v7.
- Diagnosed and resolved complex production failure classes including Oracle TCPS wallet connectivity, Azure SQL elastic pool storage constraints, SQL Server staging table race conditions, and destination schema drift.

Lead Instructor — Mar. 2023 – May 2025
Mathnasium — *Bonney Lake, Tacoma, & Lakewood, WA*

- Served as acting Center Director, managing daily operations including opening/closing and staff coordination across three locations.
- Trained new instructors on teaching methodology, conducted diagnostic assessments for prospective students, and delivered personalized K-12 math instruction targeting individual learning gaps.

Sailor — Dec. 2018 – Dec. 2022
United States Navy — *Bremerton, WA*

- Served as Mount Captain for .50 caliber gun crew of 3–4 personnel, directing target identification and engagement including live FIAC intercept.
- Executed weekly weapons maintenance procedures and 3M system documentation with zero mission failures.
- Trained in Tactical Combat Casualty Care (TCCC) for gun mount team medical response.
- Completed Naval Special Warfare preparatory training; qualified in water survival, lifesaving, and team-based operations.
- Served on Damage Control Training Team, responding to shipboard emergencies including fires and flooding.

Projects

SHA3-SHAKE Cryptographic Library | *Java*

- Implemented SHA-3 hash functions and SHAKE algorithms from scratch following FIPS 202 specification with an optimized Keccak-f[1600] permutation.
- Built a comprehensive cryptographic toolkit with a CLI interface supporting file hashing, MAC generation, and SHAKE-128 encryption/decryption, validated against NIST test vectors.

j– Compiler Extensions | *Java*

- Extended a Java subset compiler (j–) with scanner enhancements for multi-line comments and numeric literal handling, parser grammar rule additions, and AST node implementations for conditional expressions, loops, and exception handling.
- Worked across the full compilation pipeline from lexical analysis through code generation, gaining hands-on experience with language design and compiler internals.

Procedural Dungeon Generation | *Java, LibGDX*

- Implemented a Binary Space Partitioning (BSP) algorithm from scratch in Java for procedural dungeon layout generation, producing varied and replayable level structures in a 2D RPG built on LibGDX.
- Designed and built supporting gameplay systems including tile-based item enums (health potions, poison traps, bombs, pit traps), door interaction logic, and a post-dungeon player statistics screen.